Minimal-Resource Electricity

Limits, Physics and Practice: a factual, mathematical and schematic thesis on how close we can come to “unlimited” electricity from minimal matter and infrastructure, safely, efficiently and affordably

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Prepared with AI assistance (Grok, ChatGPT, Copilot). Research, calculations, figures and typesetting compiled by Grok Bot.

Edition: 1.0 · 19 October 2026 · British English

Scope: the five Stratified Source Realism (SSR) / OmniGenesis documents supplied by the author, mapped against established physics; energy densities and conversion limits; solar, wind, fission, fusion and geothermal; EROI, materials, safety and cost; a costed off-grid / community design for Peterborough, UK.

> **Core finding, stated plainly.** Infinite or perpetual electrical energy is physically impossible: a device delivering net work without an external source violates the first law of thermodynamics, and one extracting work from a single heat reservoir violates the second. What *is* achievable is electricity that is effectively unlimited *in fuel* (sunlight, wind, deuterium, breeder-reactor uranium and thorium), delivered with finite, measurable efficiency, materials, safety and cost. This document quantifies that frontier.

Labelling convention used throughout: FACT established, sourced fact · CALCULATION computed here from stated inputs · ASSUMPTION an explicit modelling assumption · SPECULATIVE unproven or future technology · PHILOSOPHICAL philosophical framing (SSR), not physics · IMPOSSIBLE forbidden by physical law.

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Abstract. The author asked for a route to infinite or unlimited electricity from minimal matter and infrastructure. Thermodynamics rules out the literal request ([1, 2, 4]). This thesis therefore does three things. First, it reads all five supplied SSR/OmniGenesis documents, extracts every energy-relevant idea, and sorts each one into established physics, speculation or philosophy. None of the documents claims a working energy source; several explicitly disclaim one. Second, it sets out the physical frontier of "electricity per unit matter". The chemical-to-nuclear jump is roughly a factor of 10^6–10^7 in specific energy, the total-annihilation ceiling is $c^2 \approx 9 \times 10^{16}$ J kg^{-1}, and the Carnot, Shockley–Queisser and Betz limits cap conversion. Fission, fusion, solar, wind and geothermal are each assessed against those limits, with current status figures (NIF target gain 4.13 in April 2025; ITER D–T from 2039; China's thorium MSR conversion in 2025). Third, it sizes and costs a real system for a medium-use UK household in Peterborough (2,500 kWh/yr), using PVGIS satellite data and October 2026 UK prices. The results: grid-tied rooftop PV pays back in about 11 years and delivers lifetime electricity at about 8–13p/kWh. Full off-grid autonomy needs roughly a 12 kWp array and a 40 kWh battery, costs more than £40,000, and works out at over £1/kWh. Shared, grid-connected community energy is the low-matter, low-cost optimum. The most nearly "unlimited" resource is the Sun, which delivers about 48,000 times humanity's average electrical power to the Earth.

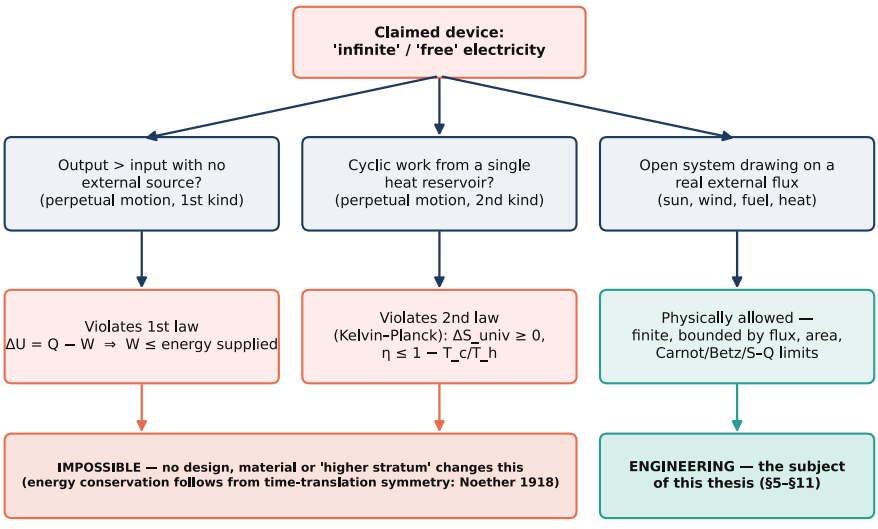


Figure 1. Decision logic: why every claimed "infinite energy" machine fails, and where real engineering begins. Schematic; the laws cited are [1, 2, 4].

1. The question, and why infinite energy is impossible

1.1 The first law: energy is conserved

FACT For any closed system, the change in internal energy equals heat supplied minus work done by the system [1, 2]:

$$\Delta U = Q - W \quad \Longrightarrow \quad \oint \mathrm{d}U = 0 \;\Rightarrow\; W_{\text{cycle}} = Q_{\text{net,cycle}}.$$

A cyclic machine that returns to its starting state therefore delivers no more work than the net heat (or other energy) fed into it. A "perpetual motion machine of the first kind", which delivers $W > 0$ with no input, contradicts this identity. Conservation of energy is not an empirical accident. Noether's theorem derives it from the invariance of physical laws under time translation [4]. If the laws are the same today as tomorrow, then

$$\frac{\partial L}{\partial t} = 0 \implies \frac{\mathrm{d}H}{\mathrm{d}t} = 0, \qquad H = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L.$$

FACT *Nuance.* In general relativity, for an expanding universe, there is in general no globally conserved total energy. Local conservation, $\nabla_\mu T^{\mu\nu} = 0$, still holds exactly [5]. No laboratory or household device can exploit cosmological expansion, so this nuance changes nothing in practice.

1.2 The second law: you cannot run on one temperature

FACT The Kelvin–Planck statement says no cyclic device can convert heat from a single reservoir entirely into work [2]. In entropy form, for a heat engine exchanging Q_h at T_h and Q_c at T_c:

$$\Delta S_{\text{univ}} = -\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \;\geq\; 0 \quad \Longrightarrow \quad \eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h} \;\leq\; 1 - \frac{T_c}{T_h}.$$

A "perpetual motion machine of the second kind", such as one that cools the ocean to power a ship, violates this inequality. Feynman's ratchet-and-pawl analysis shows why clever mechanisms do not escape it: the pawl itself fluctuates thermally [3]. Information is also physical. Erasing one bit costs at least $k_B T \ln 2 \approx 2.9 \times 10^{-21}$ J at 300 K, which has been verified experimentally [6]. A "Maxwell's demon" cannot get round the second law by bookkeeping.

> **IMPOSSIBLE** **Verdict on the literal request.** "Infinite", "perpetual", "over-unity" or "free" electrical energy is impossible for any device, at any scale, from any material or arrangement of matter. This is not a limitation of current engineering. The rest of this thesis asks the answerable question: *what is the maximum useful electricity per kilogram of matter, per square metre and per pound, under real physical law?*

2. The five source documents: summaries and verdicts

All five PDFs were converted to text (*pdftotext*) and read in full. They form one programme: Stratified Source Realism (SSR), a metaphysics in which reality (the "Omniverse") is the least fixed point of a constrained generative operator, together with its software counterpart OmniGenesis.

Document (pages)	Principal content and claims	Energy-relevant content
D1. *Stratified Source Realism: Final Competition Essay*, AI Philosophy Competition, 1st Ed.; philosophical authorship attributed to Grok (8 pp.)	The original thesis: the Omniverse is the least fixed point of a closure operator under inclusion, accessibility and a periodic ("Zodiacal") constraint. Source is identical with the operator. It argues for asymmetric recoverability, hierarchical indexicality, phenomenal character and value as sustained recoverability. It advances "no empirical claim about physical constants".	Timelines with agency must admit periodic completion so that "energy, luminous and material regularities" remain recoverable; "irreversible dissipation" would eliminate cross-temporal intervention. Energy, light and matter are "domain-relative regularities".
D2. *SSR — The Ultimate Public Edition*, v4.0, 15 Oct 2026 (18 pp.)	The most rigorous edition. It separates theorem (Knaster–Tarski) from interpretation; defines $\Gamma_C(X) = G(X) \cap \mathrm{Adm}_C$; distinguishes monotonicity, extensivity and idempotence; notes eight stratum labels (M0–M7+) against the nine claimed; formalises recurrence as $d_i(f_i(s), f_i(s')) \le \varepsilon_i$; proposes $V(h) = \int_0^T w(t) A_Q(t)\,\mathrm{d}t$.	§6: the documents "do not derive physical laws or constants from Γ_C". §5: recurrence "does not itself show that entropy is prevented from increasing". Physics' empirical work is not replaced.
D3. *SSR and OmniGenesis: Computational Realisation…*, Definitive Edition (8 pp.)	Adds $OG = \Psi(O)$, a computational-realisation map. It sets out an SSR→software correspondence (Malcolm AI Daemon, OmniGenesis Sovereign executable, policy.json) and the Sovereign–Singularity dyad, with applications in knowledge graphs, checkpointing and integrity layers. The software "does not constitute total reality".	§7.1: "continuity metrics and checkpointing protocols that preserve energy, information, and material regularities across simulation cycles".
D4. *SSR–OmniGenesis Master Codex: Ultimate Assimilation* (7 pp.)	Consolidates D2 and D3 under one equation and vocabulary, with an explicit honest boundary. Credits: originating thesis by Grok; OmniGenesis mapping with Copilot; direction and synthesis by Oliver Lux Prime Thorley.	No new energy content; inherits D1–D3.
D5. *SSR Model 4.0: Chronological Realignment Array*, 15 Oct 2026 (17 pp.)	A "conceptual engineering manual" for a time-realignment architecture. It includes ASCII schematics, a bill of materials, transfinite iteration algorithms, a Zodiacal metric filter and runtime checklists, and states that "no validated physical mechanism for a working time machine is established".	The lattice theorem "does not supply a physical actuator… an energy source, or a causal mechanism". "Zero-point recovery" is a software reset. An "Entropy Model Adapter" is required before any thermodynamic claim. Readiness is "not a physical energy or activation threshold".

Mapping the energy-relevant ideas in the five SSR/OmniGenesis documents

Idea in Oliver's theses	Established physics it touches	Verdict
Energy/light/matter as 'recoverable regularities'	Noether: conservation ⇔ symmetry	Compatible framing; no new physics
Periodic (Zodiacal) completion restores regularities	2nd law; Poincaré recurrence; Carnot cycles	A cycle restores the working fluid, not free energy
Continuity metrics 'preserve energy' across cycles	Numerical invariants; Landauer limit	Software bookkeeping, not a power source
Laws = clauses of Γ_C; cross-level constants	Empirically measured constants	Philosophy of lawhood
Least fixed point Γ_C(O)=O (Knaster–Tarski)	Steady states need energy throughput	Valid maths; no energy implication
CRA 'ignition', 'zero-point recovery'	Zero-point energy is the minimum, unextractable	Metaphor; CRA itself disclaims

Figure 2. Mapping each energy-relevant idea in the five documents to the physics it touches, with a verdict. Compiled by this report from D1–D5.

2.1 Verdicts on the energy claims

(i) "Energy, light and matter are recoverable, domain-relative regularities" (D1, D2). PHILOSOPHICAL This is compatible with physics as a *framing*. In physics, energy is the conserved charge of time-translation symmetry [4], light carries $E = h\nu$, and matter carries $E = mc^2$. SSR adds no mechanism for producing energy, and D2 says so explicitly.

(ii) Periodic ("Zodiacal") completion restores regularities that dissipation would destroy (D1). If read physically, as a cycle that undoes dissipation, this would contradict the second law IMPOSSIBLE. A thermodynamic cycle restores its *working substance*, not free energy: a Carnot engine returns to its initial state but must reject heat Q_c [2]. Poincaré recurrence exists for bounded isolated systems, but on timescales vastly longer than any engineering horizon, and it yields no work. Read conditionally, as D2 does ("a proposed stability principle"), the claim is PHILOSOPHICAL and harmless.

(iii) Checkpointing "preserves energy" across simulation cycles (D3). FACT This is valid as *software*. Numerical integrators can conserve a computed energy invariant, but that is bookkeeping, not power. The real computation consumes electricity, bounded below by Landauer's limit [6].

(iv) Laws as clauses of Γ_C; "cross-level constants" (D1, D2). PHILOSOPHICAL This is a philosophy of lawhood. Physical constants are measured, not derived, and D1 concedes that "particular physical constants … are not derived from the operator".

(v) Least fixed point $\Gamma_C(O) = O$. FACT This is valid mathematics (Knaster–Tarski), and it has no energy implication. Physical steady states (stars, organisms, power plants) persist only through continuous throughput of free energy.

(vi) CRA "ignition", "zero-point recovery" and "waveguides" (D5). FACT These are metaphors, and D5 says so itself. In physics, zero-point energy is the *minimum* energy of a quantum system. No cyclic process can extract net work from a ground state.

> **Summary.** None of the five documents asserts a source of unlimited energy. Their honest-boundary sections (D2, D4, D5) already reach the conclusion of §1. The energy question therefore has to be answered by physics and engineering, which is what the rest of this thesis does.

3. Energy per unit matter: from chemistry to E = mc²

The request for "minimal matter" is really a question about *specific energy*, $e = E/m$. Chemical reactions rearrange electrons and release a few eV per bond. Nuclear reactions rearrange nucleons and release MeV, about a million times more. The absolute ceiling is total conversion of rest mass:

$$e_{\max} = c^2 = (2.998 \times 10^8 \text{ m s}^{-1})^2 = 8.988 \times 10^{16} \text{ J kg}^{-1} \approx 25 \text{ GWh per gram.}$$

3.1 Worked nuclear specific energies

CALCULATION With $1 \text{ MeV} = 1.602 \times 10^{-13}$ J and $1 \text{ u} = 1.6605 \times 10^{-27}$ kg:

$$e_{\text{DT}} = \frac{17.59 \text{ MeV}}{(2.014 + 3.016) \text{ u}} = 3.37 \times 10^{14} \text{ J kg}^{-1}, \qquad e_{\text{D}^3\text{He}} = \frac{18.35 \text{ MeV}}{5.030 \text{ u}} = 3.52 \times 10^{14} \text{ J kg}^{-1},$$

$$e_{\text{pB}} = \frac{8.68 \text{ MeV}}{(1.007 + 11.009) \text{ u}} = 6.97 \times 10^{13} \text{ J kg}^{-1}, \qquad e_{^{235}\text{U}} \approx \frac{200 \text{ MeV}}{235.04 \text{ u}} = 8.2 \times 10^{13} \text{ J kg}^{-1}.$$

These agree with tabulated values (D–T 337 TJ kg^{-1}; uranium 80.6 TJ kg^{-1}) [7]. D–T fusion converts about 0.38% of rest mass to energy ($3.37 \times 10^{14}/c^2$); fission converts about 0.09%.

Fuel / store	Specific energy (MJ/kg)	Ratio to petrol	Status	Source
Li-ion battery (cell)	0.36–0.875	≈ 1/50	FACT storage, not a source	[7]
Wood (dry)	18	0.39	FACT	[7]
Bituminous coal	24–35	≈ 0.6	FACT	[7]
Petrol	46.4	1	FACT	[7]
Hydrogen (HHV / LHV)	141.9 / 119.9	3.1	FACT carrier, must be made	[7]
U-235, complete fission	8.06×10^7	1.7×10^6	FACT commercial since 1950s	[7]
D–T fusion	3.37×10^8	7.3×10^6	SPECULATIVE no power plant yet	[7]
Catalysed D–D (all deuterium)	5.71×10^8	1.2×10^7	SPECULATIVE far harder than D–T	[8]
Matter–antimatter annihilation	8.99×10^{10}	1.9×10^9	IMPOSSIBLE as a *source*: antimatter costs ~10^9× more energy to make than it stores	[10, 11]

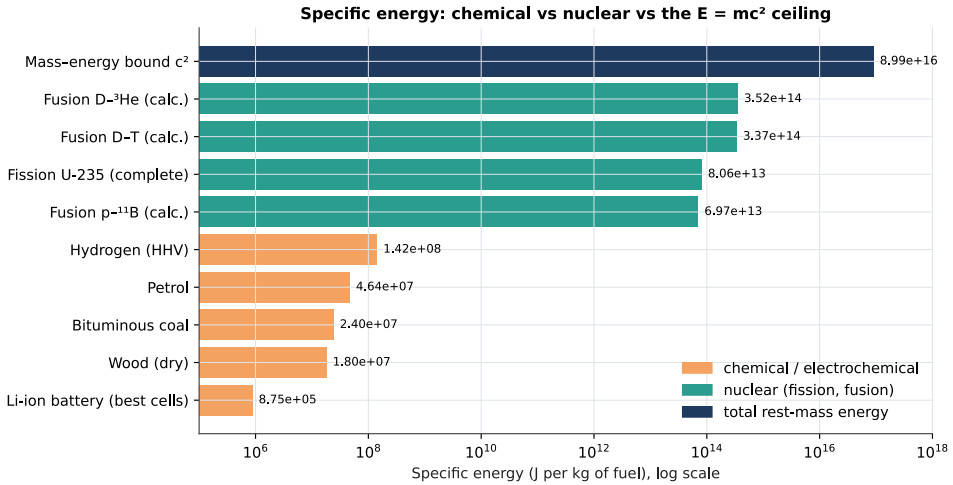


Figure 3. Specific energy by fuel class (log scale). Nuclear values calculated from reaction Q-values; chemical values from [7]. The c² bar is a bound, not an available fuel.

3.2 What “minimal matter” means for one household

CALCULATION A medium UK household now uses $E_e = 2{,}500$ kWh of electricity per year [45], which is 9.0×10^9 J. The fuel mass needed is $m = E_e/(\eta\, e)$:

$$m_{\text{coal}} = \frac{9.0\times10^{9}}{0.37\times2.4\times10^{7}} \approx 1.0\ \text{t}, \qquad m_{^{235}\text{U}} = \frac{9.0\times10^{9}}{0.33\times8.06\times10^{13}} \approx 0.34\ \text{g}, \qquad m_{c^2} = \frac{9.0\times10^{9}}{8.988\times10^{16}} = 0.10\ \text{mg}.$$

For a light-water reactor run once-through, the ENS equivalence (1 kg natural uranium ≈ 14,000 kg coal) [9] gives about 67 g of mined natural uranium per household per year ASSUMPTION (taking 29.3 MJ/kg coal-equivalent and 33% efficiency). Fast reactors, which use uranium about 60 times more efficiently [25], would bring this to about 1 g.

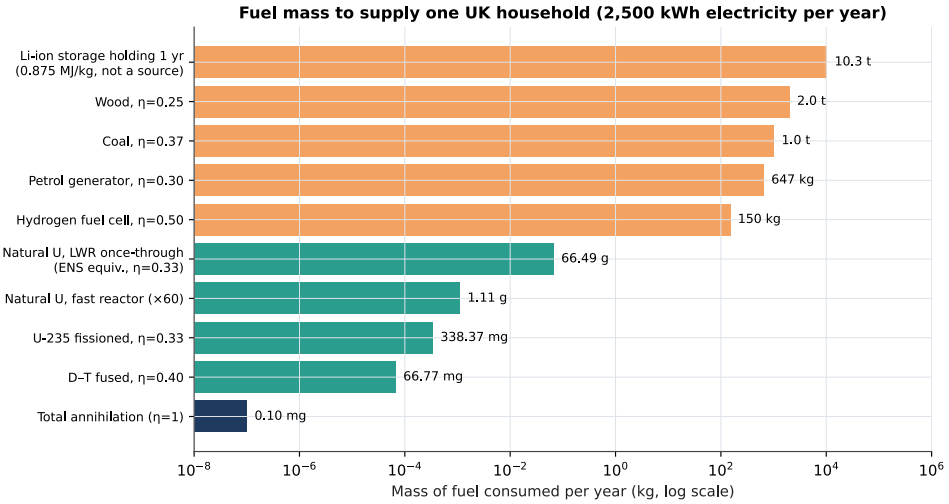


Figure 4. Fuel mass consumed per year to supply 2,500 kWh of electricity. Efficiencies η are stated assumptions; nuclear entries use the ENS equivalence [9] and the WNA fast-reactor factor [25]. Total annihilation is a theoretical lower bound only.

3.3 Deuterium: the nearest thing to inexhaustible fuel

CALCULATION Ocean water has D/H = 155.76×10^{-6} [60]. One kilogram of water contains 111.0 mol of H, so 0.0173 mol (0.0348 g) of deuterium. At 5.71×10^{14} J kg^{-1} for fully catalysed D–D [8]:

$$E_{1\ \text{kg seawater}} = 3.48\times10^{-5}\ \text{kg} \times 5.71\times10^{14}\ \text{J kg}^{-1} \approx 2.0\times10^{10}\ \text{J} \approx \text{ the heat of 430 kg of petrol.}$$

The oceans hold about 1.335×10^9 km^3 [59], roughly 1.4×10^{21} kg. Their deuterium therefore holds about 2.7×10^{31} J of fusion heat, compared with world electricity use of 1.15×10^{20} J/yr (31,920 TWh in 2025 [16]). That is a ratio of order 10^{11} years: effectively unlimited fuel. SPECULATIVE The catch: D–D fusion needs a triple product about 40 times higher than D–T (§8), and no device has achieved it.

4. Heat engines: Carnot, Rankine and the fission plant

Fission, fusion (in its D–T form), geothermal and fossil plants all make heat first and then electricity, so every one of them is bound by Carnot [2]. Real engines running at finite power are better described by the Curzon–Ahlborn efficiency at maximum power [12]:

$$\eta_{\text{C}} = 1 - \frac{T_c}{T_h}, \qquad \eta_{\text{CA}} = 1 - \sqrt{\frac{T_c}{T_h}}.$$

CALCULATION For a pressurised-water reactor steam cycle ($T_h \approx 560$ K, $T_c \approx 300$ K): $\eta_C = 46\%$ and $\eta_{CA} = 27\%$. Real LWRs achieve about 33%.

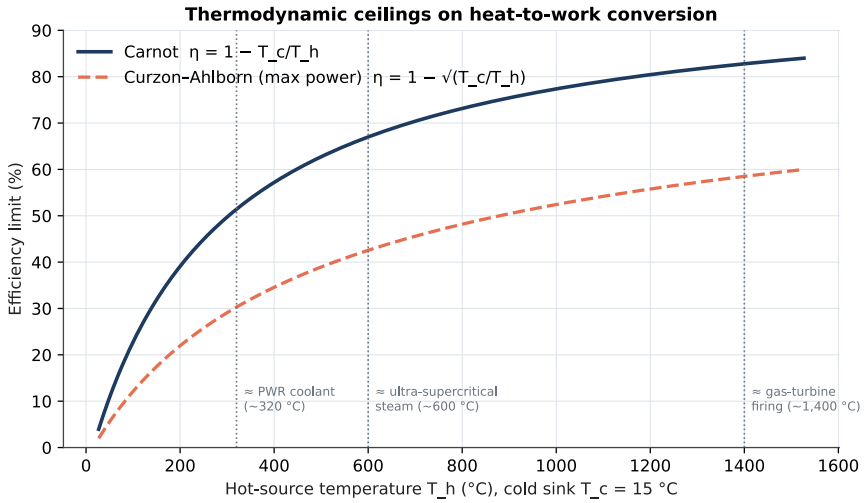


Figure 5. Carnot and Curzon–Ahlborn limits versus source temperature for a 15 °C sink [2, 12]. Temperature markers are indicative.

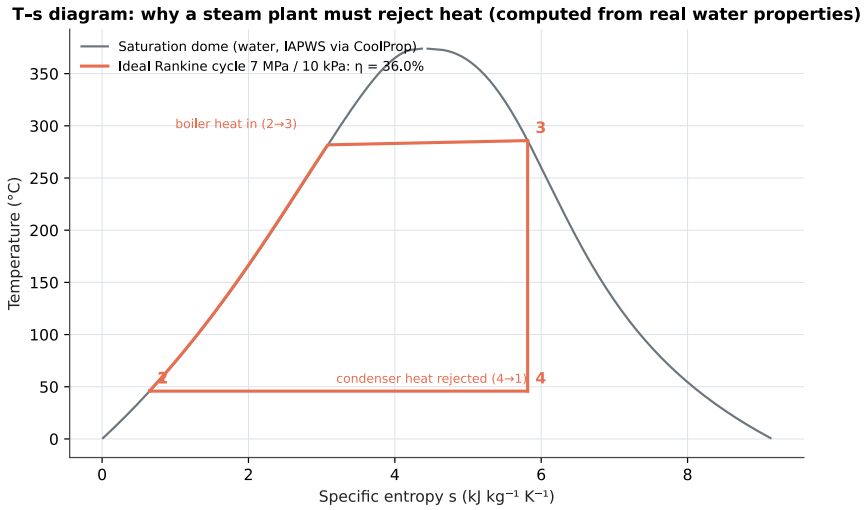


Figure 6. Temperature–entropy diagram for water with an ideal saturated Rankine cycle (7 MPa boiler, 10 kPa condenser), computed with CoolProp's IAPWS-95 implementation [13]. Even this ideal cycle reaches only about 36%. The rest is rejected at the condenser, as the second law requires.

CALCULATION The ideal cycle in Figure 6 has $\eta = [(h_3 - h_4) - (h_2 - h_1)]/(h_3 - h_2) = 36.0\%$. Every thermal plant therefore needs a cooling-water or air-cooled heat sink about twice the size of its electrical output. This is a fundamental piece of infrastructure, not a design flaw.

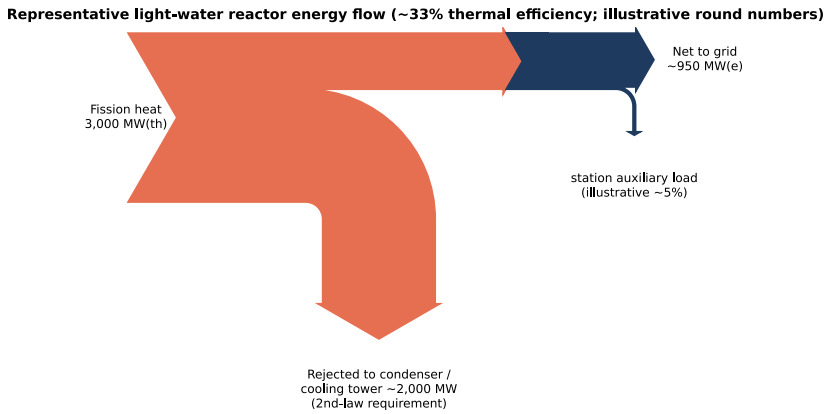


Figure 7. Representative LWR energy flow: 3,000 MW of fission heat becomes about 1,000 MW of gross electricity and about 2,000 MW of rejected heat. Round numbers are illustrative; the ~33% efficiency is typical.

5. Solar photovoltaics and the Shockley–Queisser limit

5.1 The resource

CALCULATION Total solar irradiance is 1,360.8 ± 0.5 W m^{-2} [14]. Earth's cross-section intercepts

$$P_{\oplus} = S_0\,\pi R_{\oplus}^2 = 1361 \times \pi\,(6.371 \times 10^6)^2 \approx 1.74 \times 10^{17}\ \text{W},$$

while world electricity in 2025 averaged 31,920 TWh/8,760 h = 3.64×10^{12} W [16]. That is a ratio of about 48,000. The Sun's total output is 3.828×10^{26} W [15]. Nothing else accessible to humanity comes close to "unlimited", and sunlight needs no fuel mass at all. The cost is *area, materials and storage*.

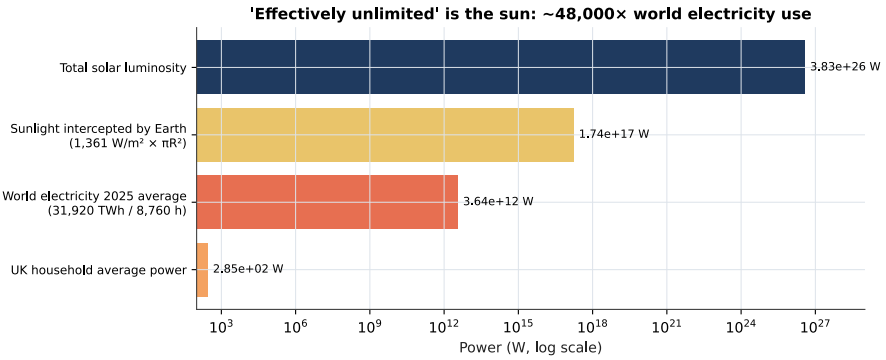


Figure 8. Power scales from one UK household to the Sun (log scale). Sources: [45, 16, 14, 15].

5.2 The detailed-balance limit

FACT A single-junction cell absorbs only photons with $E \geq E_g$. Each absorbed photon yields at most $qV < E_g$, excess energy is lost as heat (thermalisation), and the cell radiates as a blackbody at T_c [17]. The current–voltage relation under detailed balance is

$$J(V) = q\int_{E_g}^{\infty} \Phi_{\odot}(E)\,\mathrm{d}E \;-\; q\,\frac{2\pi}{c^2h^3}\int_{E_g}^{\infty} \frac{E^2\,\mathrm{d}E}{e^{(E-qV)/k_BT_c}-1}, \qquad \eta = \frac{\max_V\, VJ(V)}{\int E\,\Phi_{\odot}\,\mathrm{d}E}.$$

CALCULATION Evaluating this for the ASTM G173 AM1.5G spectrum [22] at 300 K (with the Boltzmann approximation for the dark current) gives a maximum of 33.7% at 1.34 eV (CALCULATION). Rühle's careful tabulation gives 33.16% at 1.34 eV [18]. Silicon's own limit, once Auger recombination is included, is 29.4% [19], and the best certified silicon cell reaches 27.81% [20]. Tandem (multi-junction) cells are not bound by the single-junction limit: a perovskite/silicon tandem has reached 34.85% [21].

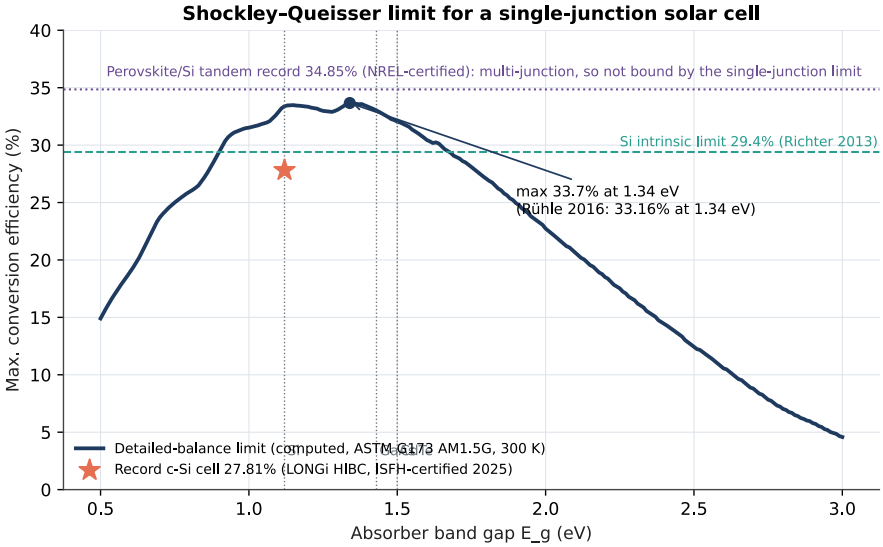


Figure 9. Shockley–Queisser efficiency versus band gap, computed here from the AM1.5G spectrum, with silicon's intrinsic limit [19] and record cells [20, 21] marked.

5.3 From sunlight to socket in Peterborough

FACT PVGIS (SARAH2 satellite data, 2005–2020) gives 1,272 kWh m^{-2} yr^{-1} in-plane irradiation and 1,035 kWh per kWp per year for a south-facing array at 38° near Peterborough, with total losses of 18.65% (angle of incidence −3.1%, spectral +1.87%, temperature and low light −4.17%, system 14%) [23]. CALCULATION With modules of about 21% efficiency, 1 kWp occupies about 4.8 m^2, so the long-run electrical power density is

$$\bar{p} = \frac{1035\ \text{kWh}}{4.8\ \text{m}^2 \times 8760\ \text{h}} \approx 25\ \text{W m}^{-2}.$$

Energy flow per 1 kWp per year, Peterborough (PVGIS-SARAH2, 38° tilt, south). Not to scale at small flows.

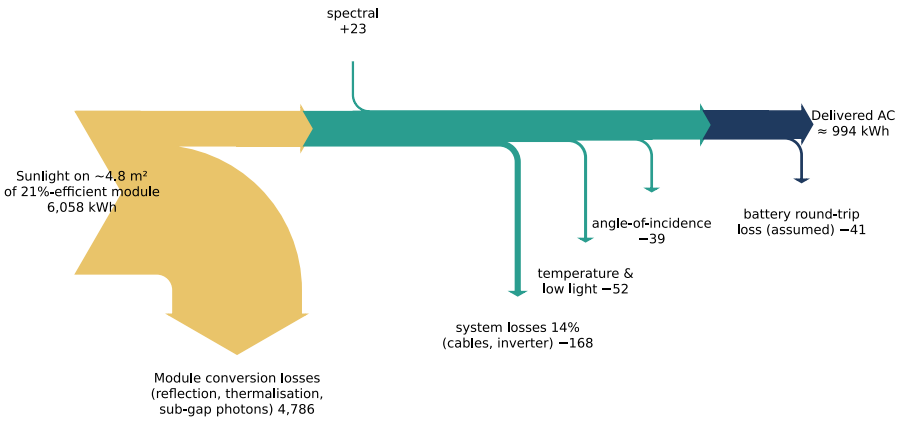


Figure 10. Annual energy flow for 1 kWp in Peterborough, using PVGIS loss factors [23]. The battery branch assumes 40% of output is cycled at 90% round-trip efficiency ASSUMPTION. Small flows are not drawn to scale.

6. Wind and the Betz limit

FACT The kinetic power crossing a rotor of area A is $\tfrac{1}{2}\rho A v^3$. Actuator-disc momentum theory, with axial induction factor a, gives [24]

$$C_p(a) = 4a(1-a)^2, \qquad \frac{\mathrm{d}C_p}{\mathrm{d}a} = 0 \Rightarrow a = \tfrac{1}{3}, \qquad C_{p,\max} = \tfrac{16}{27} \approx 0.593.$$

CALCULATION At a mean speed of 5 m s^{-1} (ρ = 1.225 kg m^{-3}), the Betz-limited power is $0.593 \times \tfrac{1}{2} \times 1.225 \times 5^3 = 45$ W per m^2 of swept area. The cubic dependence is why siting dominates everything else: halving the wind speed

cuts power eightfold. Rooftop turbines in sheltered towns perform poorly, which is why §12 rates small domestic wind as poor value.

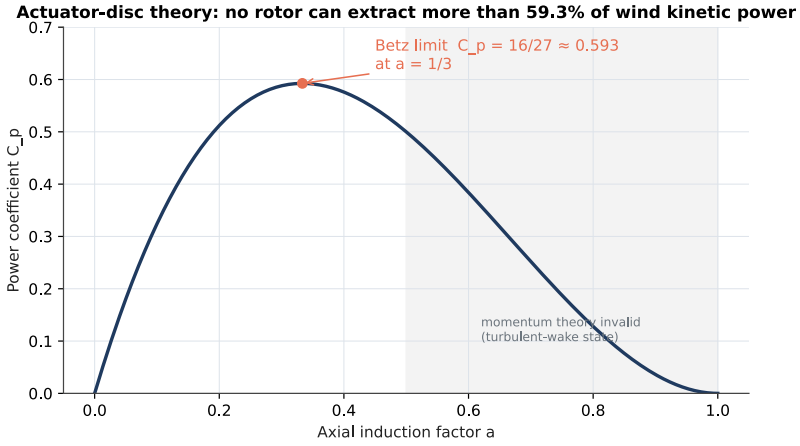


Figure 11. Power coefficient versus axial induction. No rotor, however shaped, can exceed 16/27 [24].

7. Fission: reactors, SMRs, breeders and thorium

FACT Fission is the only mature technology that delivers the nuclear ~10^6 advantage in specific energy (§3). It has one of the lowest death rates per TWh, 0.03, comparable to wind and solar [44], and the smallest bulk-material footprint per TWh of the low-carbon options (§10) [43].

Pathway	Status (verified)	Relevance to "minimal matter"
Large LWR (e.g. Sizewell C)	FACT FID 22 July 2025; ≈ £38 bn (2024 prices) [56]. Hinkley Point C CfD £92.50/MWh (2012 prices) [55]	About 67 g of natural U per household-year (§3.2); grid-delivered
SMR: NuScale US460	FACT NRC Standard Design Approval, 29 May 2025: 77 MWe per module, up to 6 modules [27]	Factory-built modules; still utility-scale
SMR: Rolls-Royce (UK)	FACT 470 MWe PWR; selected June 2025 [28]; Wylfa chosen Nov 2025 [29]; contract Apr 2026, FID expected 2029, grid mid-2030s [30]	The UK's main "small" nuclear route. Not a household option
Fast breeders	FACT use U about 60× more efficiently than thermal reactors [25]	About 1 g U per household-year; extends resources to millennia
Thorium MSR	FACT China's 2 MWt TMSR-LF1 achieved Th→U-233 conversion in 2025 [26]. SPECULATIVE commercially	Thorium is abundant; still at research scale

Why not a household reactor? Security of fissile material, licensing, shielding and decay-heat removal all scale badly downward. Fission's minimal matter is realised *through the grid*: one 470 MWe unit at a 90% capacity factor ASSUMPTION would supply $470 \times 0.9 \times 8760/2.5 \approx 1.5$ million medium households CALCULATION.

8. Fusion: Lawson, NIF, JET, ITER, STEP

8.1 The Lawson criterion

FACT A D–T plasma ignites when alpha-particle heating ($E_\alpha = 3.5$ MeV) balances losses [32]. Neglecting radiation, with $n_D = n_T = n/2$:

$$\tfrac{1}{4} n^2 \langle \sigma v \rangle E_\alpha \;\geq\; \frac{3 n k_B T}{\tau_E} \;\Longrightarrow\; n\,T\,\tau_E \;\geq\; \frac{12\,k_B T^2}{\langle \sigma v \rangle E_\alpha}.$$

The minimum falls near 14 keV. Wurzel and Hsu give $nT\tau_E \geq 2.9 \times 10^{21}$ keV m^{-3} s for ignition and 4.6×10^{20} for $Q_{\mathrm{fuel}} = 1$. D–^{3}He needs 5.2×10^{22} (18× harder), and p–^{11}B cannot ignite with $T_e = T_i$ under standard cross-sections [33].

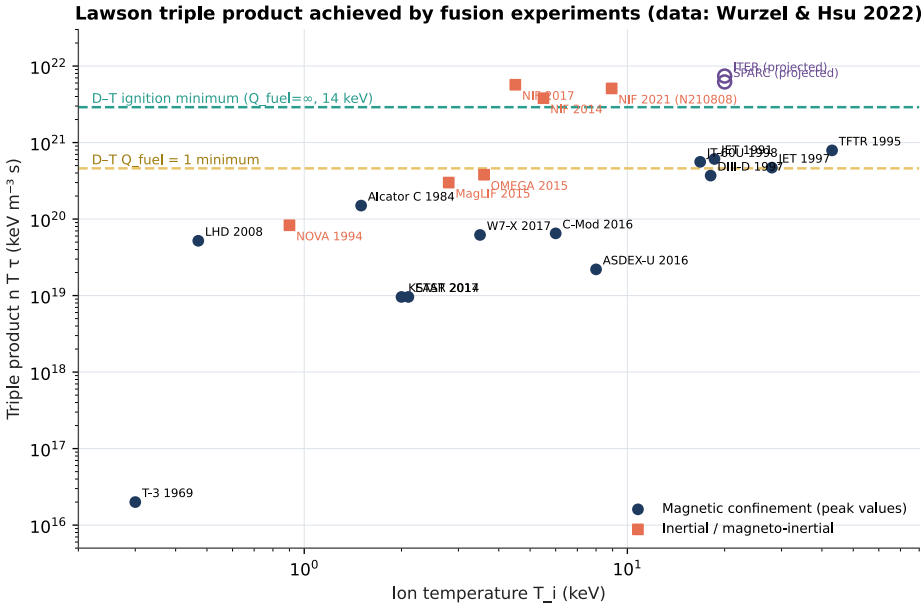


Figure 12. Triple products achieved by magnetic and inertial confinement experiments from 1969 to 2021, plus projections for SPARC and ITER. Data from Wurzel & Hsu (2022) Tables III, VI and VIII [33]. Magnetic-confinement values are $n_{i0} T_{i0} \tau_E^*$ as tabulated.

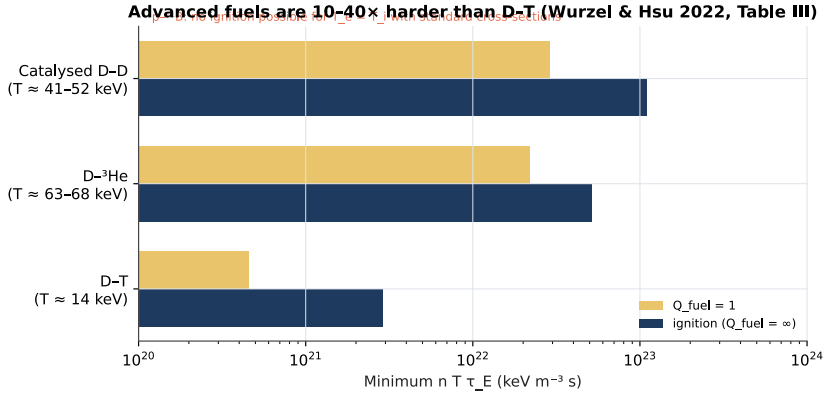


Figure 13. Minimum triple product by fuel [33]. The "aneutronic" fuels are much harder to burn than D–T.

8.2 Status of gain

Gain must be defined carefully: $Q_{\mathrm{sci}} = P_{\mathrm{fus}}/P_{\mathrm{heat}}$ (or target gain $G = E_{\mathrm{fus}}/E_{\mathrm{laser}}$), while engineering gain Q_{eng} compares net electricity out with electricity in.

- FACT **NIF.** First ignition on 5 Dec 2022: 3.15 MJ from 2.05 MJ of laser energy, $G \approx 1.5$ [35, 34]. The record on 7 Apr 2025 was 8.6 ± 0.45 MJ from 2.08 MJ, $G = 4.13$ [34]. But the lasers draw about 300 MJ from the grid per shot, so the wall-plug ratio is about 1–3% [36].
- FACT **JET.** Its final D–T campaign (late 2023) produced a record 69 MJ from about 0.21 mg of fuel, still with $Q < 1$ [38].
- FACT **ITER.** Its mission is $Q \geq 10$ (500 MW of fusion for 50 MW of heating, 400 s pulses). Under the 2024 baseline, research operations start in 2034 and D–T in 2039 [37].
- FACT **STEP (UK).** A prototype plant at West Burton targeting first operations in 2040, within a £2.5 bn fusion funding package [39]. Ministers describe fusion as "unlimited energy". Physically, it is *abundant fuel*, not unlimited energy.

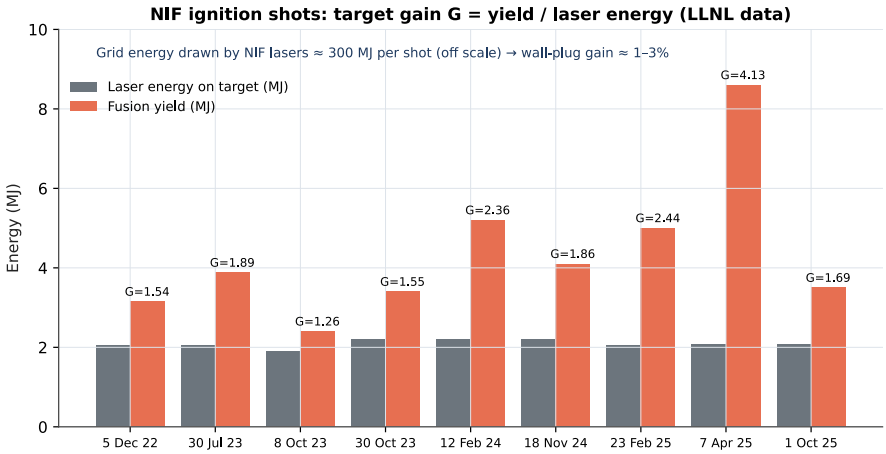


Figure 14. NIF ignition shots: laser energy on target and fusion yield [34]. The ~300 MJ drawn from the grid per shot [36] is far off the scale.

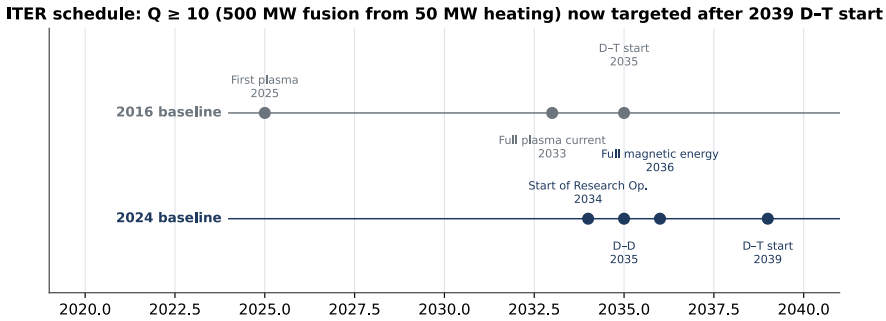


Figure 15. ITER schedule under the 2016 and 2024 baselines [37].

SPECULATIVE **Assessment.** Fusion has demonstrated scientific gain, but not engineering gain, tritium self-sufficiency or a power plant. For planning purposes it is a post-2040 grid technology with no established cost (§12).

9. Geothermal energy

FACT Global geothermal power capacity was 15.4 GW at the end of 2024 [40]. It is firm (not weather-dependent) and uses little land, but it is Carnot-limited at low source temperature. CALCULATION For a 150 °C brine and a 15 °C sink, $\eta_C = 1 - 288/423 = 32\%$; real binary-cycle plants sit well below this. DESNZ classes deep geothermal for power as a not-yet-established technology in the UK [53]. Its geothermal (high-temperature binary) materials intensity is about 5,300 t/TWh (§10) [43]. For a household, the realistic "geothermal" option is a **ground- or air-source heat pump**, which moves several units of ambient heat per unit of electricity. This does not breach the second law, because work is spent to pump heat uphill: $\mathrm{COP}_{\max} = T_h/(T_h - T_c)$.

10. System metrics: EROI, materials and safety

10.1 Energy returned on energy invested

EROI $= E_{\text{delivered}}/E_{\text{invested}}$. The net fraction available to society is $1 - 1/\text{EROI}$, so EROI 10 gives 90% net and EROI 3 gives 67%. FACT Weißbach et al. (2013), using an exergy method under German conditions, report: nuclear 75; run-of-river hydro 50 (35 buffered); coal ≈ 30; CCGT 28; CSP 21 (9.6 buffered); onshore wind 16 (4 buffered); rooftop poly-Si PV 4.0 (2.3 buffered) [41]. These values are **contested**. Murphy et al. (2022), harmonising to point of use, find PV, wind, hydro and nuclear all at or above about 10, and fossil electricity at 10–12 [42]. Both agree that the source counts for less than *storage and location*: buffering and low irradiance both lower EROI.

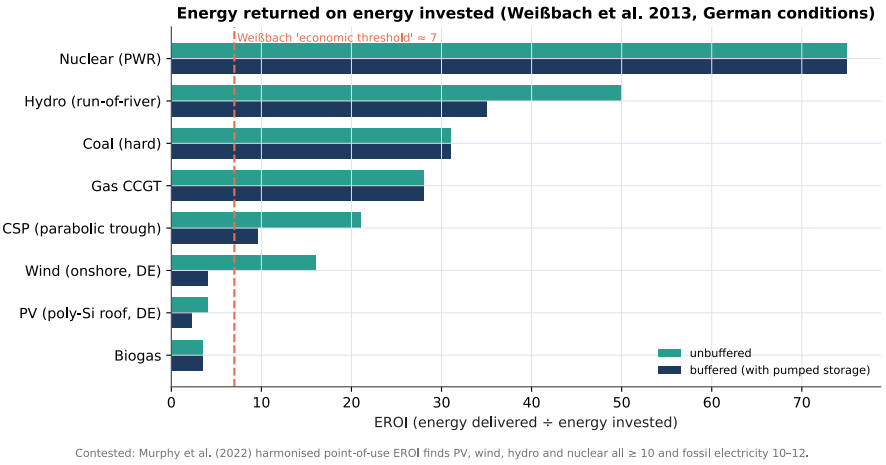


Figure 16. EROI by technology, unbuffered and with pumped-storage buffering [41], with the contrasting harmonised result of [42] noted.

10.2 Materials per TWh

FACT The US DOE gives lifetime bulk-material requirements, excluding fuel, in tonnes per TWh [43]. Nuclear PWR is about 930; wind about 10,260; silicon PV about 16,450 (mostly steel, cement and glass); hydro about 14,070. CALCULATION Per medium household-year (2.5×10^{-6} TWh), that is about 2.3 kg for nuclear, 26 kg for wind and 41 kg for PV, if the PV is grid-tied and fully used. An off-grid system oversized about fivefold (§11) multiplies the PV figure to about 200 kg per household-year, plus its battery.

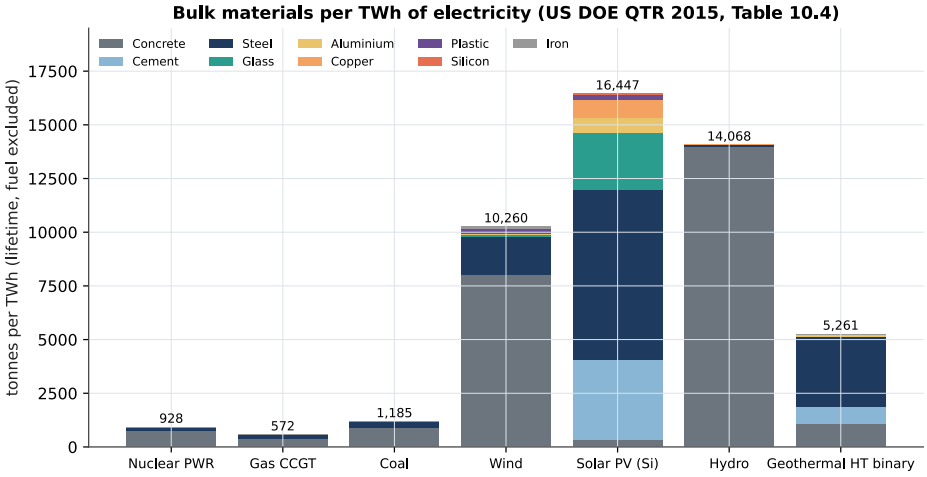


Figure 17. Bulk materials per TWh of lifetime generation, by technology (US DOE QTR 2015, Table 10.4) [43].

10.3 Safety

FACT Deaths per TWh from accidents and air pollution: brown coal 32.72, coal 24.62, oil 18.43, biomass 4.63, gas 2.82, hydro 1.30, wind 0.04, nuclear 0.03, solar 0.02 [44]. Low-carbon sources are all two to three orders of magnitude safer than fossil fuels. For a household, the dominant hazards are *electrical and fire* (DC arcs, battery thermal runaway, roof work). These are managed by certified design and installation to BS 7671 and MCS standards [61].

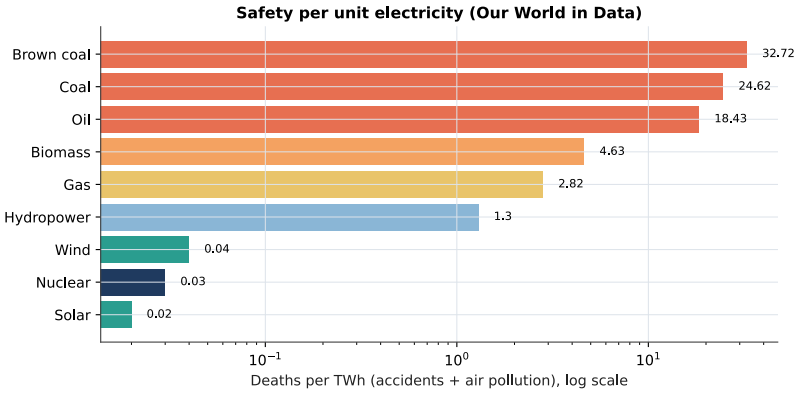


Figure 18. Death rates per unit of electricity (log scale) [44].

11. Minimal-infrastructure design for Peterborough

11.1 Demand and resource

FACT Medium UK demand is 2,500 kWh/yr [45], a daily mean of $D = 2500/365 = 6.85$ kWh. PVGIS daily yields for Peterborough range from 1.25 kWh/kWp/day in December to 3.99 in May–July (38° tilt). A steeper 60° tilt raises December output to 1.48 at a small annual cost: 995 against 1,035 kWh/kWp [23].

11.2 Sizing mathematics

CALCULATION Three sizing rules give very different answers:

$$P_{\text{annual}} = \frac{2500}{1035} = 2.4\ \text{kWp}, \qquad P_{\text{Dec}} = \frac{6.85}{1.48} = 4.6\ \text{kWp}, \qquad P_{\text{autonomy}} \approx 12\ \text{kWp} + 40\ \text{kWh}.$$

The first suits a grid-connected net-zero home. The second covers only an *average* December day. The third comes from the PVGIS off-grid simulator run over 5,844 real days, 2005–2020 [23]. Battery autonomy is $B_{\text{use}}/D = (0.8 \times 40)/6.85 \approx 4.7$ days. Dull spells longer than that defeat smaller systems:

PV / battery (60° tilt, 20% cut-off)	Days battery empty (% of all days)	Worst month	Approx. module area
6 kWp / 20 kWh	8.95%	January 39%	≈ 29 m²
8 kWp / 20 kWh	5.20%	January 24%	≈ 38 m²
10 kWp / 30 kWh	1.32%	January 9.4%	≈ 48 m²
12 kWp / 40 kWh	0.33%	January 2.8%	≈ 57 m²

CALCULATION The 12 kWp system generates about $12 \times 995 \approx 11{,}900$ kWh a year, but only 2,500 kWh is used. About 79% is curtailed, almost all of it in summer. This is the core result for the UK: **seasonal mismatch, not daily storage, drives off-grid cost and materials.**

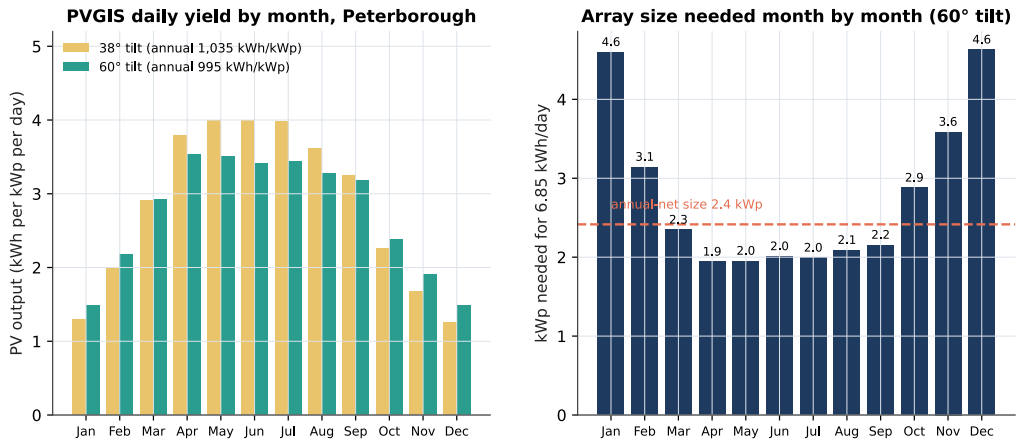


Figure 19. Left: PVGIS daily PV yield per kWp by month in Peterborough at two tilts. Right: array size needed to meet 6.85 kWh/day in each month [23].

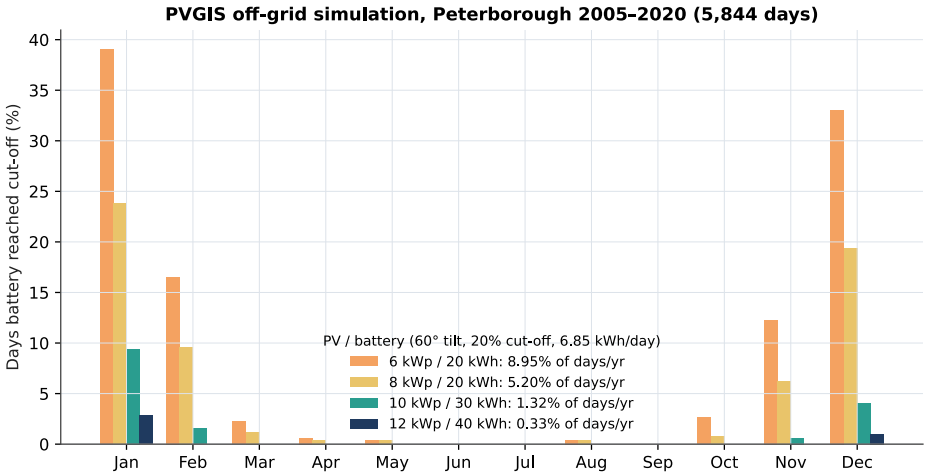


Figure 20. PVGIS off-grid simulation: share of days, by month, on which the battery reached its cut-off, for four PV/battery sizes [23].

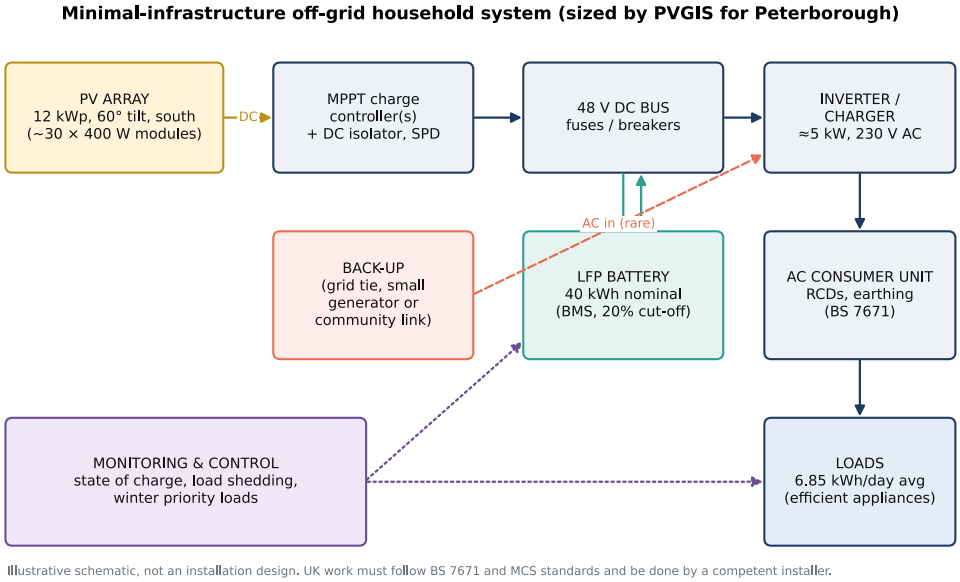


Figure 21. Minimal-infrastructure off-grid household architecture: PV array, MPPT charge control, 48 V LFP battery, inverter/charger, protected AC distribution and an optional back-up link. Illustrative schematic.

11.3 Grid-connected household and community designs

CALCULATION An hourly simulation of a 4 kWp roof, using PVGIS 2019 weather (4,298 kWh generated) and an assumed domestic load shape scaled to 2,500 kWh ASSUMPTION, gives the following. With no battery, the panels cover 37.9% of demand (1,553 kWh imported, 3,350 kWh exported). A 5 kWh battery raises this to 78.5%; a 9.8 kWh battery to 86.2% (346 kWh imported, 2,011 kWh exported). The grid acts as a free, lossless seasonal store with unlimited capacity.

CALCULATION **Community scale** (50 homes, 125 MWh/yr). About 200 kWp of shared PV (≈ 207 MWh/yr), a shared battery and one grid connection replace 50 separate 12 kWp/40 kWh off-grid systems. That needs about one third of the PV and about one tenth of the battery capacity, and it diversifies demand peaks. This is the genuinely *minimal-matter* design.

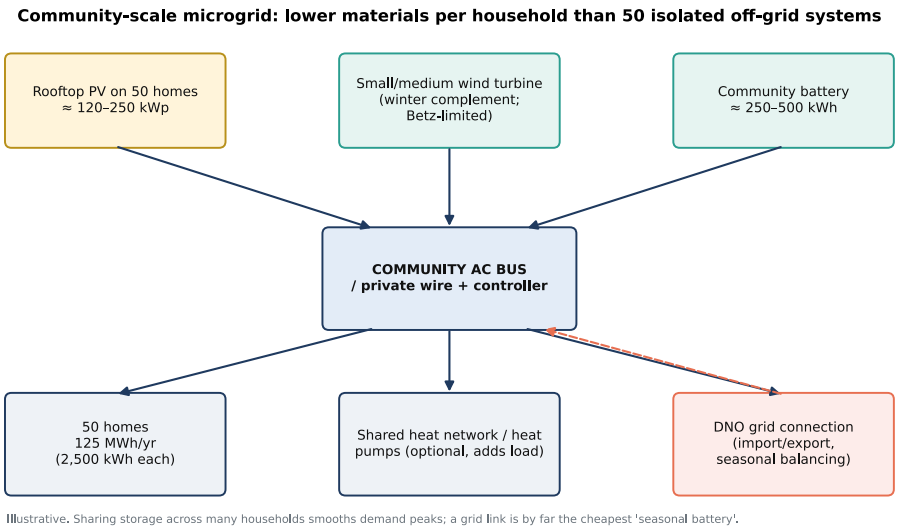


Figure 22. Community microgrid concept: shared PV, optional wind, community battery and a grid link for seasonal balancing. Illustrative schematic.

12. Costs, payback and a phased roadmap

12.1 Current UK prices (sourced and dated)

Item	Price (GB)	Date / basis	Source
Grid electricity (price cap, Direct Debit)	26.32p/kWh + 54.83p/day	1 Oct–31 Dec 2026, VAT-free period	[46]
Gas (price cap)	7.97p/kWh + 29.68p/day	Oct–Dec 2026, incl. 5% VAT	[46]
Domestic PV, installed	£1,808/kW (0–4 kW); £1,622/kW (4–10 kW); £1,267/kW (10–50 kW) mean	2025/26, MCS data, TIC basis	[47]
Home battery, installed (median)	£1,300/kWh (1–6 kWh); £890/kWh (6–11); £630/kWh (11–30)	2025/26, MCS-registered	[48]
Air-source heat pump	≈ £13,431 median, minus £7,500 BUS grant	Q4 2025 median; BUS 2026	[50, 49]
Small wind, 6 kW pole-mounted	≈ £35,000; saves ≈ £650/yr	EST guidance (accessed Oct 2026)	[51]
Export (Smart Export Guarantee)	12p/kWh Outgoing Octopus; 16p (4–7 pm) / 9p Prime; 4.1p export-only SEG	since 1 Mar 2026	[52]
Large nuclear	Sizewell C ≈ £38 bn; HPC CfD £92.50/MWh (2012 £)	FID July 2025	[56, 55]
SMR (Rolls-Royce)	No public strike price; reported estimates ≈ £2–3 bn per unit SPECULATIVE	FID expected 2029	[30, 31]
Fusion	No commercial cost exists SPECULATIVE; STEP within £2.5 bn programme	first operations target 2040	[39]

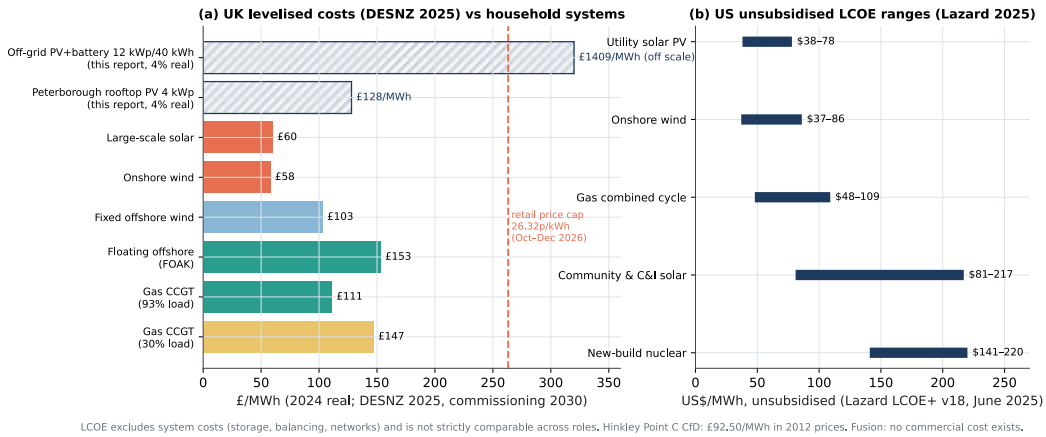


Figure 23. (a) DESNZ levelised costs for projects commissioning in 2030 (2024 real £/MWh; CCGT values read from DESNZ Figure 5) [53], compared with this report's household LCOEs and the retail price cap [46]. (b) Lazard unsubsidised US ranges [54].

12.2 Worked household cases (Oct 2026 prices held constant)

ASSUMPTION Assumptions: 25-year life; one £1,000 inverter replacement in year 12; 0.5%/yr PV degradation in the LCOE calculation; 4% real discount rate; for the off-grid case, battery replacement at year 15 at today's price.

Case	Capex	Annual benefit	Simple payback	Lifetime cost of electricity
A 4 kWp PV, grid-tied, Outgoing 12p	£7,232	£249 import saved + £402 export = £651	11.1 yr	12.8p/kWh (4%) · 8.4p (undiscounted)
A′ as A, export at 4.1p	£7,232	£249 + £137 = £386	18.7 yr	as A
B A + 9.8 kWh battery	£15,952	£567 + £241 = £808	19.7 yr (battery alone ≈ 55 yr)	—
C off-grid 12 kWp + 40 kWh	≥ £40,404 (excl. off-grid inverter)	£858 (whole bill incl. standing charge)	≈ 47 yr, so never within battery life	£1.41/kWh (4%) · £1.07 (undiscounted)
D community share, 4 kWp at 10–50 kW price	£5,068	≈ £651 if export terms match A	7.8 yr	9.3p/kWh (4%)
E 6 kW small wind	£35,000	£650 (EST)	≈ 54 yr	≈ £1.04/kWh (4%, 20 yr)
F heat pump replacing gas boiler	£5,931 after BUS	Running cost ≈ neutral: 2,850 kWh × 26.32p ≈ £750 vs 9,500 kWh gas × 7.97p ≈ £757, + £108 gas standing charge if disconnected ASSUMPTION SCOP 3, boiler 90%	—	main benefit: decarbonised heat

CALCULATION The arithmetic for Case A: self-use $= 2500 - 1553 = 947$ kWh × £0.2632 = £249; export 3,350 kWh × £0.12 = £402; payback $7232/651 = 11.1$ yr. For Case C: $40{,}404/858 = 47$ yr. Both use the hourly simulation in §11.3. Subsidy and tariff changes (export rates, VAT, standing charges) can shift these results by several years.

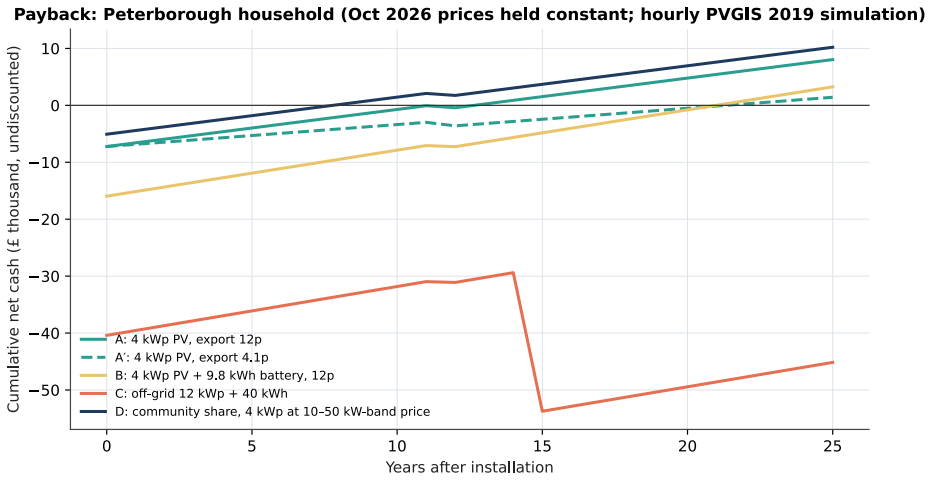


Figure 24. Undiscounted cumulative cash position for the household cases. Off-grid autonomy (C) never pays back; shared community PV (D) pays back fastest.

12.3 Phased roadmap, ranked by cost per kWh and feasibility

Phase	Action (in rank order)	Indicative cost per kWh	Feasibility
Now (0–2 yr)	1. Efficiency: LED, efficient appliances, insulation	lowest (energy not used)	FACT proven
	2. 3–4 kWp rooftop PV, MCS-installed, best SEG tariff	8–13p	FACT proven
	3. Heat pump with BUS grant when the boiler is due	≈ cost-neutral to run	FACT proven
	4. Battery only with time-of-use/Prime export spreads	marginal	FACT proven, weak economics
≈ 5 yr	5. Community energy: shared PV, community battery	≈ 9p (est.)	FACT needs organisation
	6. Buy grid power from utility solar and onshore wind	£58–60/MWh generation cost [53]	FACT
	7. Demand flexibility, smart EV charging	reduces system cost	FACT
≈ 20 yr	8. SMRs / Sizewell C on the grid	Lazard US $141–220/MWh today [54]	SPECULATIVE cost risk
	9. Enhanced geothermal	not established in UK	SPECULATIVE
	10. Fusion (STEP 2040 prototype)	none established	SPECULATIVE not cost-effective today
	✗ "Free energy" / over-unity devices	infinite (they do not work)	IMPOSSIBLE

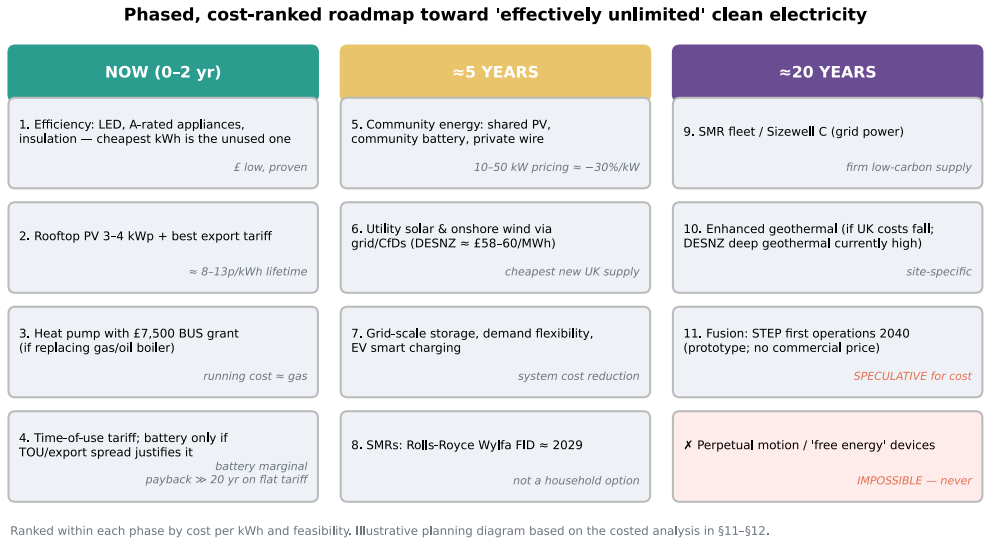


Figure 25. Phased roadmap toward effectively unlimited clean electricity, ranked within each phase by cost and feasibility. Illustrative planning diagram based on §12.

13. What not to buy: free-energy scams

IMPOSSIBLE **Never buy** any device marketed as a "free-energy generator", "magnetic motor", "over-unity", "zero-point", "self-running", "fuel-less generator", "quantum energy" or "perpetual" power source. By §1, a box that outputs more energy than it receives from an identifiable external source cannot exist. Demonstrations that appear to work are hiding an input: a concealed battery, a mains feed, mismeasured AC power, or a too-short test.

Other common traps. (1) Plug-in "power-saving" boxes: household meters bill real energy, so capacitors or "power-factor" gadgets do not lower domestic kWh bills. (2) "Free solar" or "government grant" cold calls and

doorstep sales; verify any scheme on GOV.UK or Ofgem and never give bank details under pressure [57]. (3) Headline “fully installed” solar prices that omit material conditions; the ASA has upheld complaints against such adverts [58]. (4) Small wind turbines on sheltered urban roofs (§6, Case E).

Safe buying checklist. Use an MCS-certified installer, which is also required for SEG eligibility [52]; get three written quotes and compare £/kW with the DESNZ medians in §12.1 [47]; ask for a PVGIS-style yield estimate; and check the electrical work meets BS 7671 [61].

14. SSR as a philosophical frame for energy

PHILOSOPHICAL This section is explicitly philosophical. It shows how SSR's stratification can *frame* energy physics without replacing it or adding to it. SSR orders domains M0–M7+ by asymmetric recoverability: higher strata recover lower regularities by structure-preserving restriction (D1, D2). Physics also has a hierarchy of effective theories separated by energy scale. Chemistry (eV) is recoverable from atomic physics, nuclear physics (MeV) from QCD, and so on up to the Planck scale. Read in this light, an SSR "restriction map" resembles the physicist's notion of an effective field theory valid below a cutoff. Each stratum's "cross-level constants" resemble the parameters that survive coarse-graining.

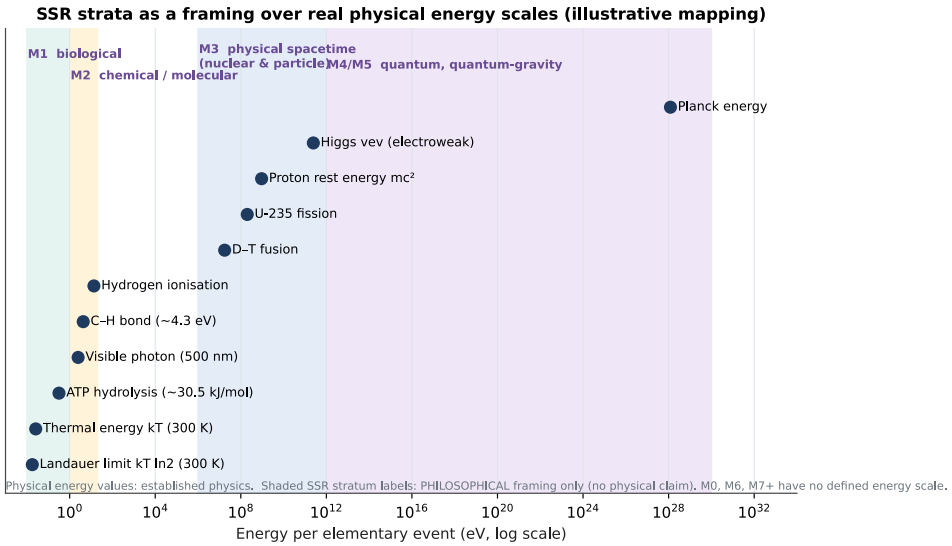


Figure 26. Physical energy scales per elementary event (established physics) with SSR stratum labels overlaid as a philosophical framing only. M0, M6 and M7+ correspond to no defined energy scale.

This framing yields three disciplined lessons:

1. **Energy flows downward in usefulness.** Each stratum's accessible energy per event rises with the stratum. That is why moving from M2 (chemical) to M3 (nuclear) gives the factor of 10^6 in §3. No stratum offers an exemption from conservation.
2. **Periodic completion, read physically, is a thermodynamic cycle.** It restores state variables while exporting entropy. Recoverability is paid for with free energy, which is a fair statement of what an engine, an organism or a battery does.
3. **Value as sustained recoverability** (D1, D2) maps naturally onto *resilience*: a household that can recover its energy service through a dull January (§11) by sharing with its community and the grid embodies the principle better than an isolated, oversized system.

Nothing in this section is a physical claim. SSR's Ultimate Public Edition itself insists that the framework "does not replace physics' empirical work" (D2).

15. Conclusions

1. IMPOSSIBLE Infinite or perpetual electricity is impossible: the first and second laws, grounded in Noether symmetry, forbid it. None of the five SSR/OmniGenesis documents claims otherwise.
2. FACT The real frontier per unit of matter is nuclear. Fission already delivers about 0.34 g of U-235 per household-year. Fusion of seawater deuterium would make the fuel effectively inexhaustible (SPECULATIVE for power).
3. FACT The real frontier per unit of infrastructure and cost today is solar and wind, bounded by the Shockley–Queisser and Betz limits and delivered through the grid at about £58–60/MWh for new UK

utility projects.

4. CALCULATION For a Peterborough household, a 3–4 kWp grid-tied PV system pays back in about 11 years at about 8–13p/kWh. Full off-grid autonomy needs about 12 kWp and 40 kWh, costs more than £40,000 and works out at over £1/kWh. Community energy with a grid link is the minimal-matter, minimal-cost optimum.
5. SPECULATIVE SMRs and fusion are grid-scale, post-2030 and post-2040 options respectively. Neither is cost-effective today, and neither is a household device.
6. PHILOSOPHICAL SSR can frame this landscape as a stratified hierarchy of recoverable regularities, as a lens rather than a power source.

References

All URLs accessed 19 October 2026. Computations performed for this report: specific energies, fuel masses, the Shockley–Queisser curve (pvlib ASTM G173 spectrum), the Rankine cycle (CoolProp), the hourly self-consumption model and the LCOE/payback model. PVGIS API queries were run live on 19 October 2026.

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Credits

Author: Oliver Lux Prime Thorley, originator of the request and of the SSR/OmniGenesis programme analysed in §2 and §14. **AI assistance:** Grok (xAI), ChatGPT (OpenAI) and Copilot (Microsoft), as credited in the source documents; research synthesis, verification against the sources above, calculations, figures and typesetting compiled by Grok Bot. Figures were generated with Matplotlib, CoolProp and pvlib; equations were typeset with KaTeX. Schematic diagrams are marked as illustrative. All numerical data are sourced or computed from stated inputs.